

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		0.26		1		1		1
		(ii)		$\frac{0.52 + 0.54}{2} \text{ (1)}$ $= 0.53 \text{ [ignores anomaly] so Tom is correct (1)}$ Alternative: $\frac{0.52 + 0.54 + 0.26}{3} \text{ (1)}$ $[= 0.44] \text{ but this includes the anomaly so Tom is correct (1)}$ Conclusion must be present to award 2 marks			2	2	2	2
	(b)			Selecting a pair of cases with doubling e.g. 1 to 2 or their corresponding speeds (1) if speed doubles it should be e.g. 1.7 to 3.4 or 2.3 <u>isn't double</u> 1.7 (1) [so Tom is wrong] Alternative: Speed doubles from 1.7 to 3.4 (1) but cake cases don't double from 1 to 2 or it increases from 1 to 4 (1) [so Tom is wrong]			2	2		2
	(c)	(i)		$\text{uncertainty} = \frac{0.48 - 0.44}{2} \text{ (1)}$ $= 0.02 \text{ [s] (1)}$	1	1		2	2	2
		(ii)		Human error in timing / reaction time	1			1		1
				Question 5 total	2	2	4	8	4	8